# Crane Control

## State Machine

| State | Position | Load | Next State (condition) | Notes |
|---|---|---|---|---|
| 0 | PowerUp State.<br>We can only be sure of position if one of the lower limit switches is active | | 4 (Lower Limit Switch 0)<br>12 (Lower Limit Switch 1)<br>else 6 | If position is unknown, initiate upward and guess we are in position 0, however we could be in position 1, but system will self correct.<br>Relies on being able to safely perform an upward winch even if the winch is already 'up' |
| 1 | 0 | Up | 2 after time delay provided Limit Switch 0 is OK<br>else 8 | Random time delay |
| 2 | 0 | Downward | 3 (Lower Timeout)<br>4 (Lower Limit Switch 0)<br>12 (Lower Limit Switch 1) | |
| 3 | 0 | Down 0 Timeout | 4 (Lower Limit Switch 0)<br>12 (Lower Limit Switch 1) | Error Condition. Stay here until a limit switch gets triggered |
| 4 | 0 | Down 0 Confirmed | 6 (random time delay) | Random time delay |
| 5 | 0 | Down 0 Error | 7 (fixed transit delay) | Limit switch 0 stuck on, set error state |
| 6 | 0 | Upward | 7 (fixed transit delay)<br>5 (Lower Limit Switch 0) | |
| 7 | 0 | Up | 8 (random time delay) | Random time delay |
| 8 | Rotate Clockwise | Up | 9 (fixed transit delay) | |
| 9 | 1 | Up | 10 after time delay.<br>Provided Limit Switch 1 is OK | Random time delay |
| 10 | 1 | Downward | 11 (Lower Timeout)<br>4 (Lower Limit Switch 0) | |
| 11 | 1 | Down 1 Timeout | 4 (Lower Limit Switch 0)<br>12 (Lower Limit Switch 1) | Error Condition. Stays here until a limit switch gets triggered |
| 12 | 1 | Down 1 Confirmed | 14 (random time delay) | Random time delay |
| 13 | | Down 1 Error | 15 (fixed transit delay) | Limit Switch 1 stuck on, set error state |
| 14 | 1 | Upward | 15 (fixed transit delay) | |

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|---|---|---|---|---|
| | | | 13 (Lower Limit Switch 1) | |
| 15 | 1 | Up | 16 | Random time delay |
| 16 | Rotate AntiClockwise | Up | 1 (fixed transit delay) | |